8. The active site of an enzyme is generally a

a) cleft
b) Indentation
c) Hole
d) Tube

9. Which one of the following conditions is least likely to denature an enzyme?

a) a high temperature
b) an extreme pH
c) heavy metal ions
d) a low temperature

10. According to lock and key & induced fit models which of the following is not true?

a) Induced fit model is more specific than lock and key model
b) In lock and key model the active site is more RIGID than induced fit model
c) In induced fit model the active site can undergo conformational change
d) In lock and key model, the enzyme is considered as a lock and substrate is considered as a key

11. Which of the following is the best description of an enzyme?

a) they allow chemical reactions to proceed very quickly
b) they increase the rate at which a chemical reaction approaches equilibrium
c) they make a reaction thermodynamically favorable
d) all of the above e. none of the above

12. Given an enzyme with a K_m = 10m M and V_{max} = 100 m mol/min. If [S] = 100 m M, which of the following will be true?

a) A 10 fold increase in V_{max} would increase velocity 10 fold y
b) A 10 fold decrease in K_m would increase velocity
c) Both (a) and (b)
d) A 10 fold increase in V_{max} would decrease velocity 20 fold.

13. The relationship between K_m and V_{max} is known as

a) Haldane equation
b) Michaelis Menten equation
c) Numerical solution approach
d) Gibbs-Helmholtz equation

14. A competitive inhibitor of an enzyme is usually

a) a highly reactive compound
b) a metal ion such as Hg^{2+} or Pb^{2+}
c) structurally similar to the substrate.
d) water insoluble

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